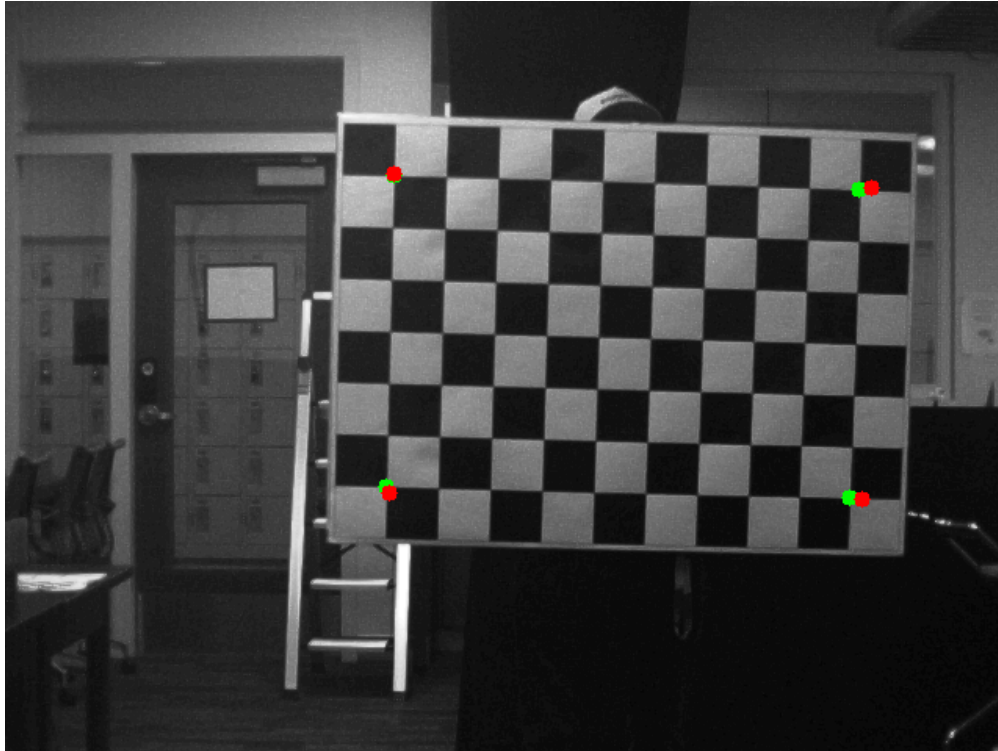


3D Reconstruction and Trajectory Est.

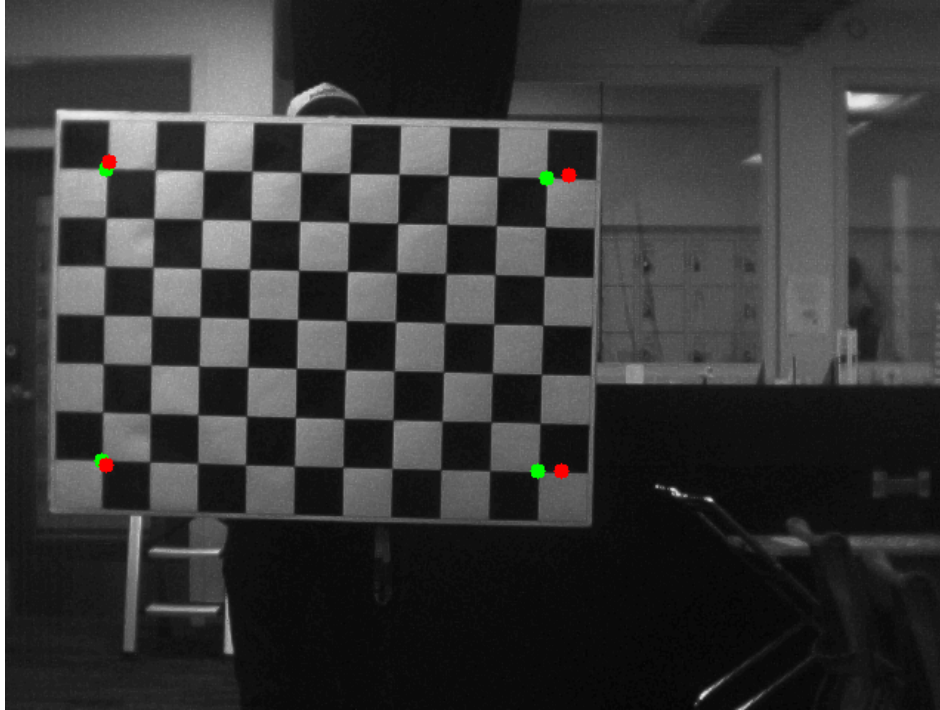
Allan Howe Ecen 631 Robotic Vision - Mar/3/2026

Task 1

The left image with green drawing the chessboard original frame, and the red drawing the new undistorted and rectified point location:



And the right:



Here are the points:

3D points w.r.t LEFT camera:

```
[[-34.378746 -13.56506 190.45781 ]
 [ 1.0801594 -12.91176 196.91788 ]
 [-35.04519 10.269925 191.82443 ]
 [ 0.346676 11.084156 198.4121  ]]
```

3D points w.r.t RIGHT camera:

```
[[-16.2592723 -15.38787213 187.2679125 ]
 [ 19.1131016 -14.73072064 194.18575667]
 [-16.99165189 8.43110097 188.86116214]
 [ 18.31173437 9.24778479 195.90731756]]
```

Rigid transform verification error:

```
[0.00000000e+00 2.84217094e-14 1.77635684e-15 0.00000000e+00]
```

Task 2

This task I did the perspective function in python. See code.

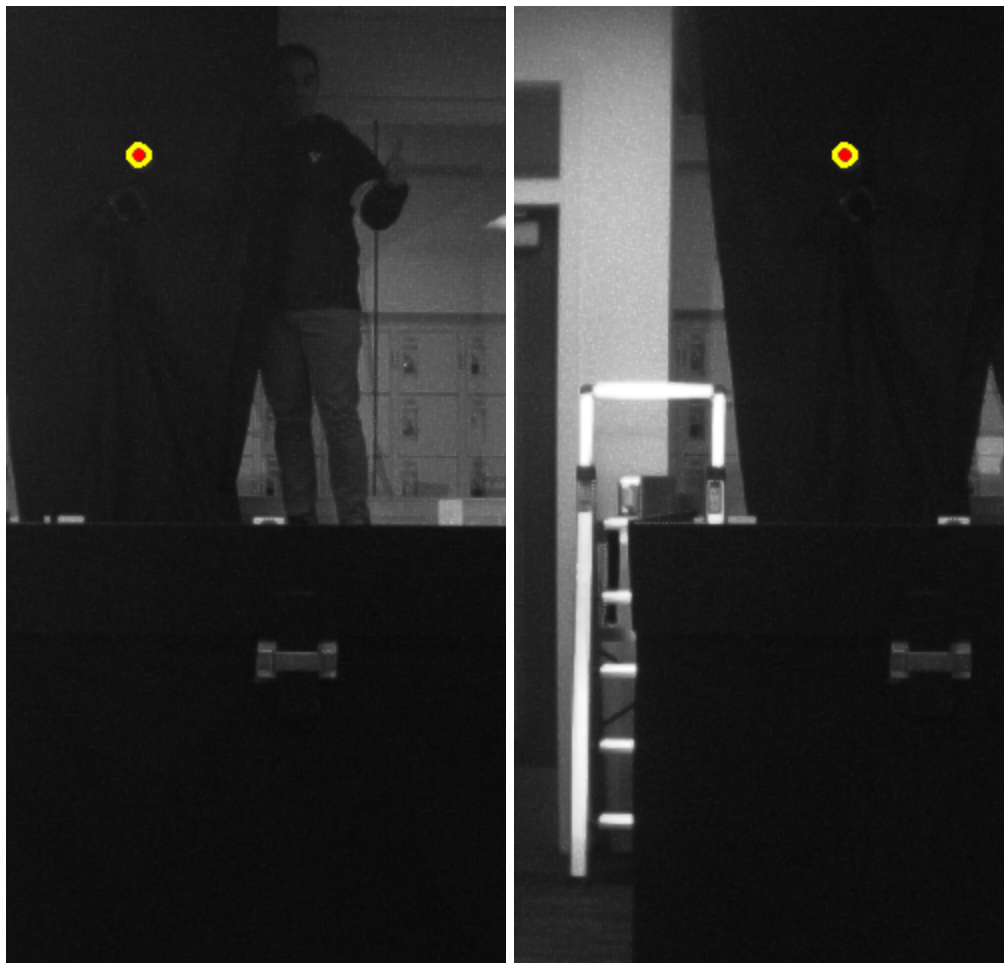
Task 3

This task tracks the ball. I improved my detection from the first lab.

0th image



5th image



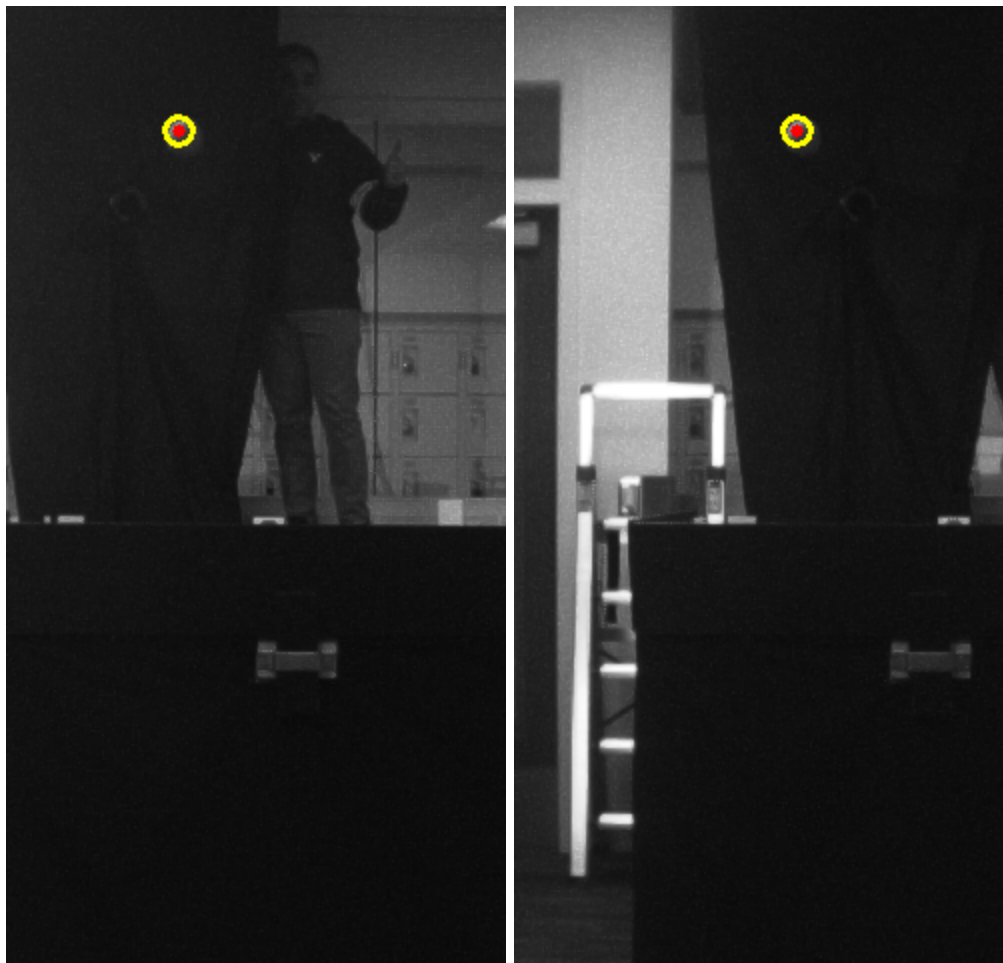
10th image pair:



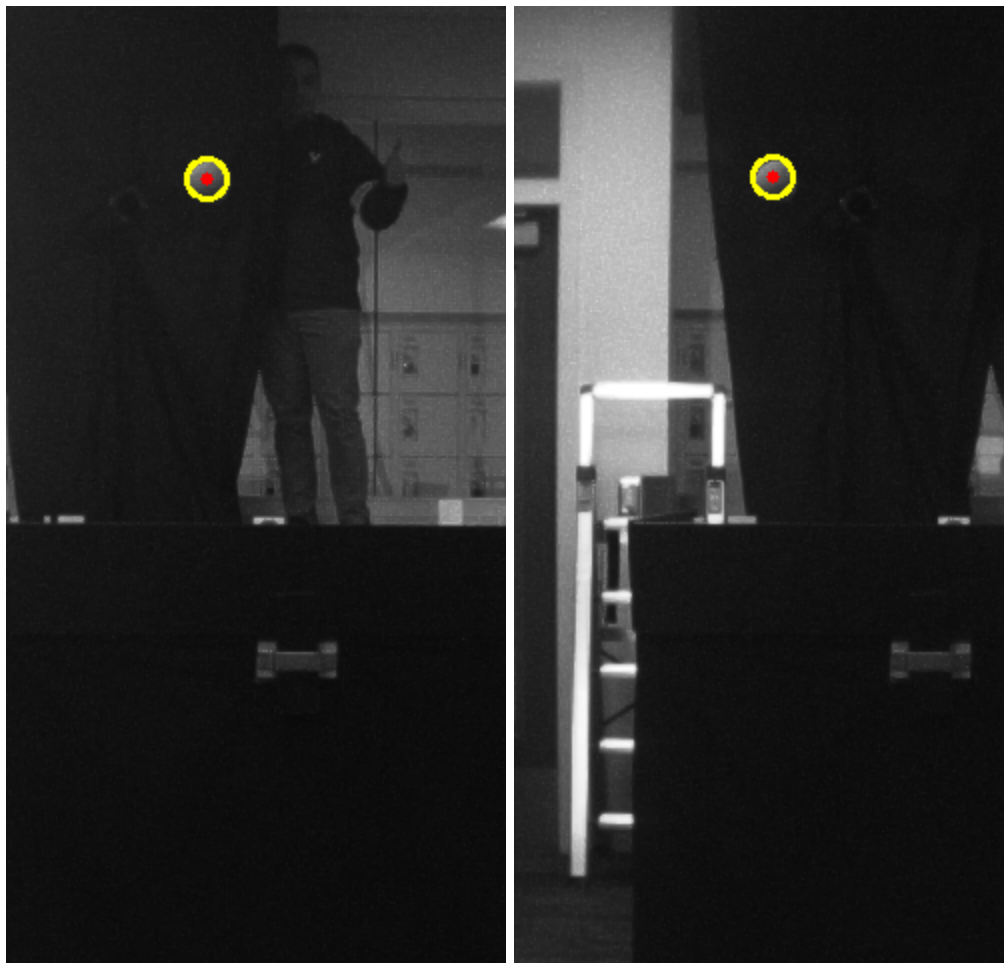
15th image



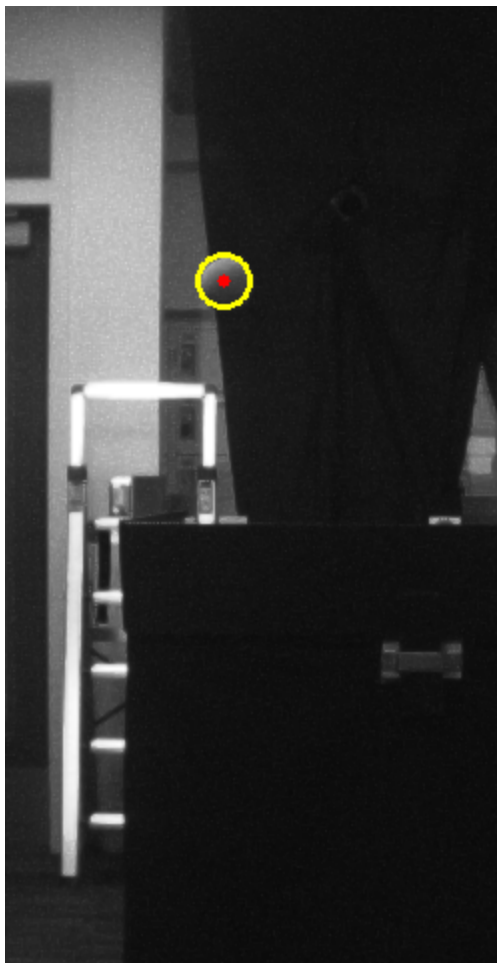
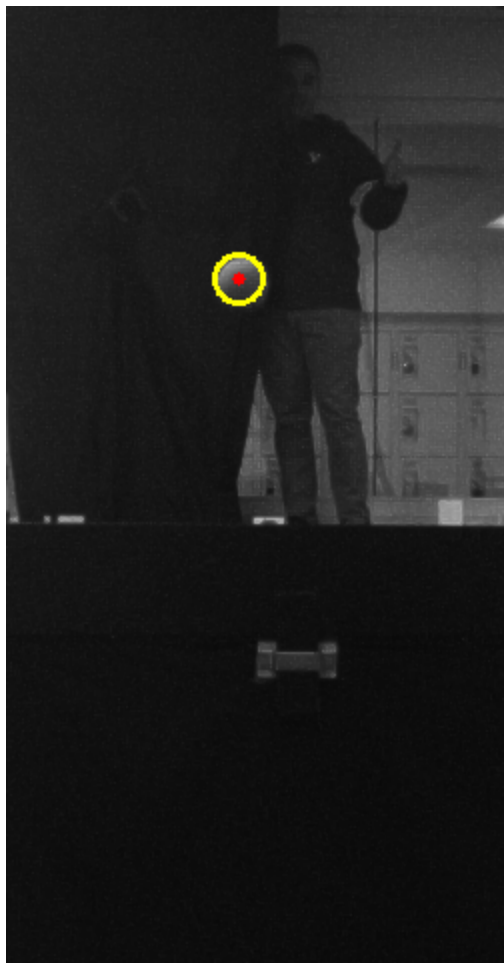
20th image pair:



25th image pair:

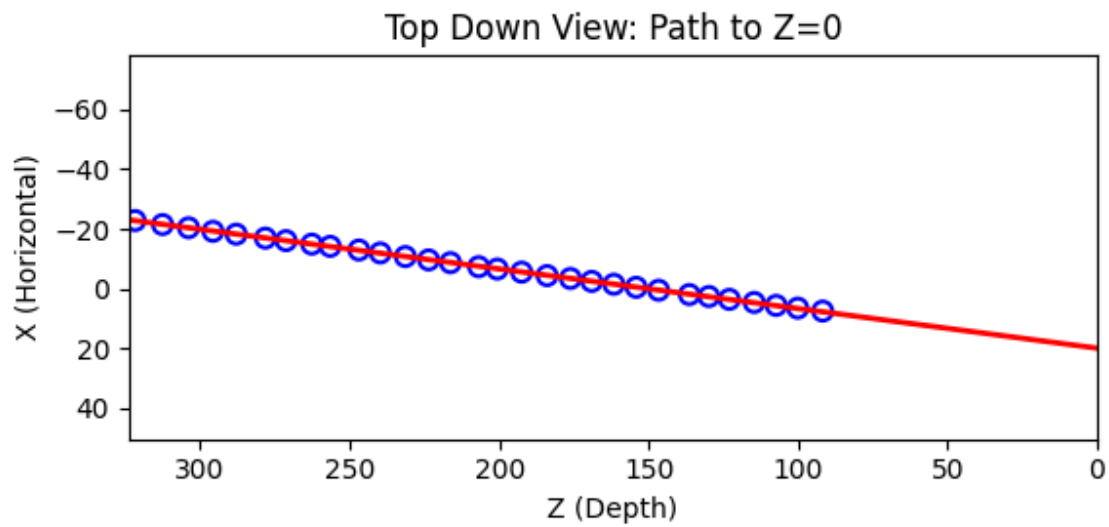


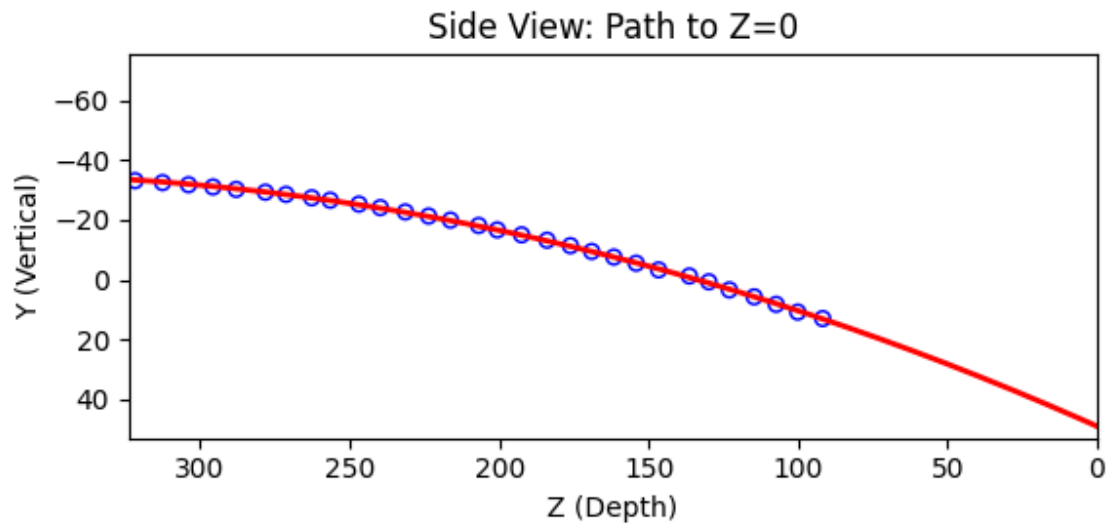
30th image pair:



Task 4

For this task I used a polynomial fit for the y, since gravity is parabolic. And I used a straight line for X, since drift is usually linear. The X and Y values are the line of best fit when $Z=0$.





Predicted intercept at Z=0:
X = 19.868095863038324
Y = 49.04675269022748